Rileigh **Bandy**

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Education

Ph.D., Computer Science *2024*

The University of Colorado, Boulder

- Advisor: Rebecca Morrison, Ph.D.
- Dissertation: *Uncertainty Representations in White- and Black-Box Models: Quantifying Model-Form and Measurement Errors in Computational Science*

M.S., Computer Science *2022*

The University of Colorado, Boulder *3.97/4.0 GPA*

B.S., Computer Science *2019*

The University of Texas at Austin *3.67/4.0 GPA*

Research and Industry Experience

BAE Systems, Inc. *Broomfield, CO*

Senior Systems Engineer - Space & Mission Systems *Apr. 2026 - Present*

Sandia National Laboratory *Albuquerque, NM*

Postdoctoral Appointee *Jun. 2024 - Apr. 2026*

Research and Development Intern *May 2022 - Jun. 2024*

- Developed modular, object-oriented Python software integrating machine learning (Jax) with physics-based models for data assimilation (ensemble Kalman filters), surrogate modeling, and reliability improvements.
- Collaborated with a multidisciplinary team to enhance reduced gas–surface chemistry models, applying machine learning-based uncertainty quantification and validation methods for tractable and accurate hypersonic flight simulations.
- Integrated physics-based models with data-driven components to fuse accelerometer sensor data, applying frequency-domain signal processing (FFT) to enable robust state estimation in dynamics systems.

University of Colorado *Boulder, CO*

Research Assistant *Aug. 2019 - May 2024*

- Developed distribution-based uncertainty representations for real-time space weather state estimation, enabling probabilistic predictions from deterministic machine learning models in PyTorch.
- Collaborated with an interdisciplinary team to develop and deliver software using machine learning and MPI for parallel computation, which remains in operational use.
- Designed and optimized C++ stochastic enrichment operators to improve predictive accuracy of reduced-order physics-based models, deploying large-scale simulations on supercomputers using MPI, Slurm, and automated workflows.

University of Texas at Austin *Austin, TX*

Research Fellow *May 2016 - Aug. 2019*

- Contributed to the open-source Transition State Atomic Simulation Environment (TSASE) Python library, enhancing global optimization methods for computational chemistry and materials applications. (http://theory.cm.utexas.edu/tsase/)
- Developed and maintained a database to facilitate collaborative sharing of simulation results.

Institute of Pure and Applied Mathematics *Los Angeles, CA*

Research Experience for Undergraduates Participant *Jun. - Aug. 2018*

- Collaborated with industry sponsors, a graduate mentor, and undergraduate students to apply data science and machine learning techniques in Python and MATLAB to simulate additive manufacturing processes.

Electric Reliability Council of Texas *Taylor, TX*
CYBERSECURITY INTERN *May - Dec. 2017*

- Improved the company's security posture through the creation of an automated Open-Source Intelligence program that alerts security analysts of threats to the company or its personnel on the Clearnet and dark web.
- Created educational phishing exercises to improve company-wide security awareness.

Awards & Distinctions

2025	**Selected Participant**, Rising Stars in Computational and Data Sciences	*Austin, TX*
2024	**Recipient**, Clive F. Baillie Memorial Scholarship	*Trieste, Italy*
2023	**Recipient**, Space Weather with Quantified Uncertainties Student Travel Fellowship	*Cambridge, MA*
2023	**Winner of the Best Paper in Model Validation and Uncertainty Quantification**, Conference Proceedings of the Society for Experimental Mechanics (SEM) Series	*Austin, TX*
2020	**Recipient**, Dean's Summer Research Fellowship	*Boulder, CO*
2018	**Winner of the Student E-Poster Competition in the Technology, Engineering, and Math category**, the American Association for the Advancement of Science (AAAS) annual meeting	*Austin, TX*

Journal and Conference Papers

7. **Rileigh Bandy**, Enrico Camporeale, Andong Hu, Thomas Berger, and Rebecca Morrison. Accurate and reliable uncertainty estimates for deterministic predictions: extensions to under and overpredictions. 2026. URL: https://doi.org/10.48550/arXiv.2604.08755
6. Rebekah White, **Bandy, Rileigh**, and Teresa Portone. Inference in the presence of model-form uncertainties: Leveraging a prediction-oriented approach to improve uncertainty characterization. 2026. URL: https://doi.org/10.48550/arXiv.2601.04396
5. **Rileigh Bandy**, Rebecca Morrison, Erin Mussoni, and Teresa Portone. Hybrid physics-data enrichments to represent uncertainty in reduced gas-surface chemistry models for hypersonic flight. 2025. URL: https://doi.org/10.48550/arXiv.2509.08137
4. **Rileigh Bandy**, Rachel Washington, Rebecca Morrison, and Teresa Portone. Isolating and quantifying uncertainties in the vibration isolation round-robin challenge. In *Proceedings of the IMAC XLIII Conference*, Orlando, FL, USA, January 2025
3. **Rileigh Bandy**, Teresa Portone, and Rebecca E. Morrison. Stochastic model correction for the adaptive vibration isolation round-robin challenge. In *Proceedings of the IMAC XLII Conference*, Orlando, FL, USA, January 2024. URL: https://doi.org/10.1007/978-3-031-68893-5_8
2. **Rileigh Bandy** and Rebecca Morrison. Stochastic model corrections for reduced Lotka-Volterra models exhibiting mutual, commensal, competitive, and predatory interactions. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 34(1), 2024. URL: https://doi.org/10.1063/5.0159043
1. **Rileigh Bandy** and Rebecca Morrison. Quantifying model form uncertainty in spring-mass-damper systems. In *Proceedings of the IMAC XLI Conference*, Austin, TX, USA, February 2023. Springer. URL: https://doi.org/10.1007/978-3-031-37003-8

Presented Work

- 13 conference presentations; 6 poster presentations

Service and Professional Activities

Students (Co-)Mentored

- Paul Smith, M.S. (2025)
- Eric Pabon Cancel, Ph.D. student (2025)
- Rachel Washington, M.S. student (2024)
- Mayah Drayton, B.S. student (2024)

Graduate Mentor *2020 - 2022*

Access and Inclusion Peer Mentoring Program

- Served as a mentor for first-year undergraduate underrepresented minority students in Engineering.
- Met regularly with mentees to provide support as they transitioned to college.

Post Secondary Volunteer Tutor *2021*

Boulder "I Have a Dream" Foundation

- Tutored post-secondary students in STEM subjects.

Institutional Service

- Co-Organizer, Sandia UQ Working Group

Conference Minisymposia (Co)-Organized

- SIAM Conference on Uncertainty Quantification (UQ26)
- SIAM Conference on Computational Science and Engineering (CSE25)
- 18th U.S. National Congress on Computational Mechanics (USNCCM18)

Reviewer for Journals

- Chaos: An Interdisciplinary Journal of Nonlinear Science
- Review of Scientific Instruments

Technical Skills

Programming Languages	Python, Julia, Bash, C++, Java, R, MATLAB, SQL, JavaScript, HTML.
Software Development	Git, Agile, Amazon Web Services, Docker.
High Performance Computing	MPI, OpenMPI, Slurm.